① Problemă curs

$$x^{(1)} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \quad x^{(2)} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \quad x^{(3)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$y^{(1)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad y^{(2)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad y^{(3)} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$\bar{V}(x) = \text{ReLu}(x) = \max(0, x)$

$n = 3$ (3 exemple), $d = 3$ (3 linii X), $q = 2$ (features) ↘ linii Y

Presupunem $h = 2$

$$W^{(1)}_{\in \mathbb{R}^{d \times h}} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix} \in \mathbb{R}^{3 \times 2} \qquad b^{(1)}_{\in \mathbb{R}^{1 \times h}} = \begin{pmatrix} 0 & 1 \end{pmatrix} \in \mathbb{R}^{1 \times 2}$$

$$W^{(2)}_{\in \mathbb{R}^{h \times q}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2 \times 2} \qquad b^{(2)}_{\in \mathbb{R}^{1 \times q}} = \begin{pmatrix} -1 & 1 \end{pmatrix} \in \mathbb{R}^{1 \times 2}$$

$$X = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$ → $x^{(1)T}$

nr parametrii = $3 \times 2 + 1 \times 2 + 2 \times 2 + 1 \times 2 = 14$

$$A = X W^{(1)} + b^{(1)} = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -1 & 2 \\ 2 & -1 \\ 1 & -1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ 2 & 0 \\ -1 & 0 \end{pmatrix}$$

$$H = \text{ReLu}(A) = \begin{pmatrix} 0 & 3 \\ 2 & 0 \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$

$$O = H W^{(2)} + b^{(2)} = \begin{pmatrix} 0 & 3 \\ 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} -1 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ -1 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -4 & 1 \\ -1 & 3 \\ -1 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$

$$\mathcal{L}(w) = \frac{1}{m} \sum_{i=1}^{m} \frac{1}{2} (\hat{y}^{(i)} - y)^2$$

$$= \frac{1}{2m} \text{Tr} \left[(O-Y)^T (O-Y) \right]$$

$Y^{(1)^T}$

$$O - Y = \begin{pmatrix} -4 & 1 \\ -1 & 3 \\ -1 & 1 \end{pmatrix} - \begin{pmatrix} \boxed{0 \quad 1} \\ 1 \quad 2 \\ 2 \quad 3 \end{pmatrix} = \begin{pmatrix} -4 & 0 \\ -2 & 1 \\ -3 & -2 \end{pmatrix}$$

$$\mathcal{L}(W^{(1)}, b^{(1)}, W^{(2)}, b^{(2)}) = \frac{1}{6} \operatorname{Tr}\left(\begin{pmatrix} -4 & -2 & -3 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} -4 & 0 \\ -2 & 1 \\ -3 & -2 \end{pmatrix} \right) =$$

$$= \frac{1}{6} \operatorname{Tr}\begin{pmatrix} 29 & 4 \\ 4 & 5 \end{pmatrix} = \frac{34}{6} = \frac{17}{3}$$

$L = \frac{1}{2m} \operatorname{Tr}\left((O-Y)^T (O-Y)\right)$ → pt ex 10

$R = \frac{\lambda}{2} \left(\|W^{(1)}\|_F^2 + \|W^{(2)}\|_F^2 \right)$ $\quad \|W\|_F = \sqrt{\operatorname{Tr}(W^T W)}$

$J = L + R$

$$\frac{\partial J}{\partial W^{(2)}} = \frac{\partial (L+R)}{\partial W^{(2)}} = \underbrace{\frac{\partial L}{\partial W^{(2)}}}_{\text{I}} + \underbrace{\frac{\partial R}{\partial W^{(2)}}}_{\text{II}}$$

I. $$\frac{\partial L}{\partial W^{(2)}} = \frac{\partial L}{\partial \theta} \cdot \frac{\partial \theta}{\partial W^{(2)}} = \frac{\partial L}{\partial \theta} \frac{\partial (W^{(2)} h)}{\partial W^{(2)}} = \frac{\partial L}{\partial \theta} h^T$$

$h^T \cdot \frac{\partial L}{\partial \theta}$

$2 \times 3 \quad 3 \times 2$

- $$\frac{\partial L}{\partial \theta} = \frac{\partial}{\partial \theta} \left(\frac{1}{2m} \operatorname{Tr}\left((O-Y)^T (O-Y)\right) \right) = \frac{1}{m} (O-Y)$$

$$\Rightarrow \frac{\partial L}{\partial W^{(2)}} = \frac{1}{m} h^T (O-Y)$$

II. $$\frac{\partial R}{\partial W^{(2)}} = \lambda W^{(2)}$$

$$\Rightarrow \frac{\partial J}{\partial W^{(2)}} = \frac{1}{m} H^T (O-Y) + \lambda W^{(2)} \in \mathbb{R}^{h \times 2}$$

$$\frac{\partial J}{\partial w^{(1)}} = \frac{\partial (L+s)}{\partial w^{(1)}} = \underbrace{\frac{\partial L}{\partial w^{(1)}}}_{\text{I}} + \frac{\partial s}{\partial w^{(1)}}$$

(I) $$\frac{\partial L}{\partial w^{(1)}} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w^{(1)}} = \frac{\partial L}{\partial z} \frac{\partial (w^{(1)} x)}{\partial w^{(1)}} = x^T \frac{\partial L}{\partial z} \in \mathbb{R}^{d \times h}$$

$d \times m \quad m \times h$

- $$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial h} \cdot \frac{\partial h}{\partial z} = \frac{\partial L}{\partial h} \cdot \frac{\partial \phi(z)}{\partial z} = \frac{\partial L}{\partial h} \odot \phi'(z)$$

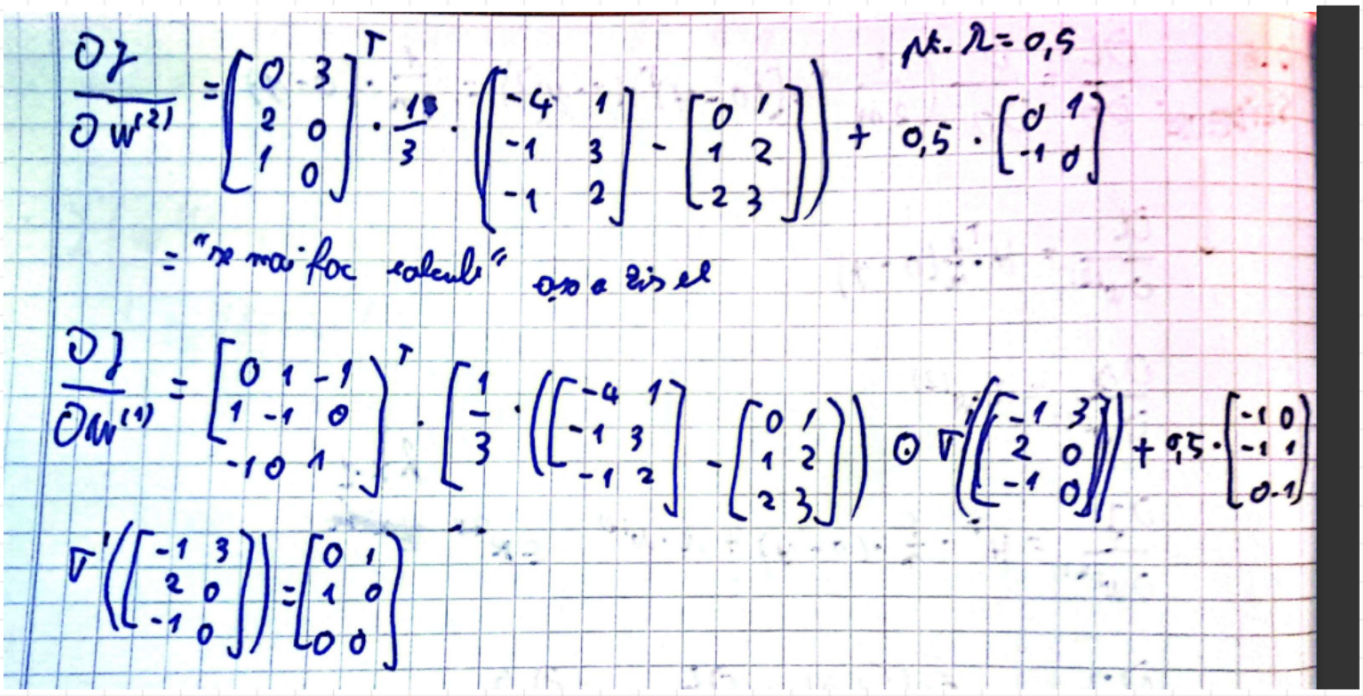
$$\frac{\partial L}{\partial h} = \frac{\partial L}{\partial \theta} \cdot \frac{\partial \theta}{\partial h} = \frac{\partial L}{\partial \theta} \cdot \frac{\partial (w^{(2)} h)}{\partial h} = \frac{\partial L}{\partial \theta} w^{(2)T} \in \mathbb{R}^{m \times h}$$

$m \times q \quad q \times h$

$$= \frac{1}{m} (\theta - y) \, w^{(2)T}$$

(II) $$\frac{\partial s}{\partial w^{(1)}} = \lambda w^{(1)}$$

$$\frac{\partial J}{\partial w^{(1)}} = x^T \cdot \left(\frac{1}{m} (\theta - y) w^{(2)T} \right) \odot \phi'(z) + \lambda w^{(1)}$$



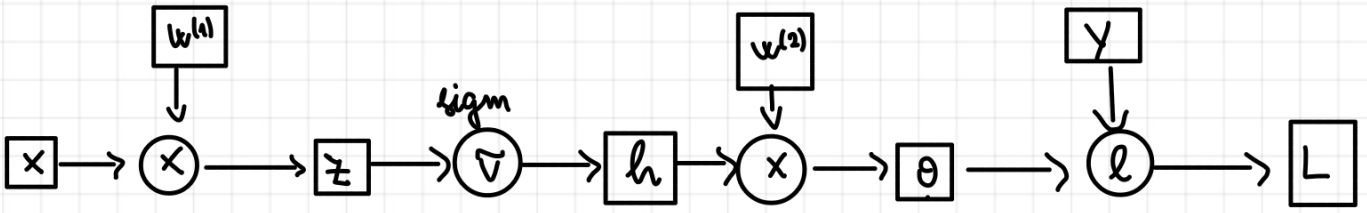
$$Tr(A^T \cdot A) = a_{11}^2 + a_{12}^2 + \ldots + a_{23}^2 = \|A\|_F^2$$

(2.)

compute the output of attention pooling for scaled dot-product attention.

10. Define an MLP for regression, having 3 inputs, one hidden layer of 2 units, with the sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ activation function, and 1 output. We ignore bias terms for simplicity. Moreover, the loss function is the squared error loss, and is denoted as L. Compute symbolically $\frac{\partial L}{\partial \boldsymbol{W}^{(1)}}$. Note that the first intermediate variable is $\boldsymbol{z} = \boldsymbol{X}\boldsymbol{W}^{(1)}$ and not $\boldsymbol{z} = \boldsymbol{W}^{(1)}\boldsymbol{x}$, which means that you will have to deduce the equations in this new setting. Numerical application: $\boldsymbol{W}^{(1)} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \\ 1 & 0 \end{bmatrix}, \boldsymbol{W}^{(2)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \boldsymbol{X} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \boldsymbol{y} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

$n = 2 \qquad d = 3 \qquad h = 2 \qquad q = 1$



$$Z = X W^{(1)} \in \mathbb{R}^{2\times 2}$$

$$h = \sigma(Z) \in \mathbb{R}^{2\times 2} \qquad \sigma(z) = \frac{1}{1+e^{-z}}$$

$$\theta = h W^{(2)} \in \mathbb{R}^{2\times 1}$$

$$\ell(\theta, y) = \frac{1}{2m} \mathrm{Tr}\left((\theta - y)^T (\theta - y) \right)$$

$$\frac{\partial L}{\partial W^{(1)}} = \frac{\partial L}{\partial Z} \cdot \frac{\partial Z}{\partial W^{(1)}} = \frac{\partial L}{\partial Z} \cdot \frac{\partial (X W^{(1)})}{\partial W^{(1)}} = \underbrace{X^T}_{d\times m} \underbrace{\frac{\partial L}{\partial Z}}_{m\times h}$$

$$\frac{\partial L}{\partial Z} = \frac{\partial L}{\partial h} \cdot \frac{\partial h}{\partial Z} = \frac{\partial L}{\partial h} \frac{\partial \sigma(Z)}{\partial Z} = \frac{\partial L}{\partial h} \odot \sigma'(Z)$$

$$\frac{\partial L}{\partial h} = \frac{\partial L}{\partial \theta} \cdot \frac{\partial \theta}{\partial h} = \frac{\partial L}{\partial \theta} \cdot \frac{\partial (h \cdot W^{(2)})}{\partial h} = \frac{\partial L}{\partial \theta} W^{(2)T}$$

$$\frac{\partial L}{\partial \theta} = \frac{\partial}{\partial \theta} \left(\frac{1}{2m} \mathrm{Tr}\left[(\theta - y)^T (\theta - y) \right] \right) = \frac{1}{2m} \cdot 2 (\theta - y) = \frac{1}{m} (\theta - y)$$

$$\frac{\partial L}{\partial W^{(1)}} = X^T \left[\left(\frac{1}{m} (\theta - y) \cdot W^{(2)T} \right) \odot \sigma'(\underbrace{XW^{(1)}}_{Z}) \right]$$

$$Z = X W^{(1)} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 0 & -1 \end{pmatrix}$$

$$h = \begin{pmatrix} \frac{1}{1+e^{-2}} & \frac{1}{1+e^{1}} \\ \frac{1}{1+e^{0}} & \frac{1}{1+e^{1}} \end{pmatrix} = \begin{pmatrix} 0,88 & 0,27 \\ 0,5 & 0,27 \end{pmatrix}$$

$$o = h\, w^{(2)} = \begin{pmatrix} 0,88 & 0,27 \\ 0,5 & 0,27 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0,27 \\ 0,27 \end{pmatrix}$$

$$o - y = \begin{pmatrix} 0,27 \\ 0,27 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -0,73 \\ 1,27 \end{pmatrix}$$

$$(o-y)\, w^{2T} = \begin{pmatrix} -0,73 \\ 1,27 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -0,73 \\ 0 & 1,27 \end{pmatrix}$$

$$\sigma'(z) = \left(\frac{1}{1+e^{-z}} \right)' = \left(1+e^{-z}\right)' = -1 \left(1+e^{-z}\right)^{-2} \frac{d}{dz}\left(1+e^{-z}\right) =$$

$$= \frac{e^{-z}}{\left(1+e^{-z}\right)^2} = \left(1 - \sigma(z)\right) \odot \sigma(z)$$

$$\sigma(x\, w^{(1)}) = \sigma(x\, w^{(1)}) \odot \left(1 - \sigma(x\, w^{(1)})\right) =$$

$$= \begin{pmatrix} 0,88 & 0,27 \\ 0,5 & 0,27 \end{pmatrix} \odot \begin{pmatrix} 0,12 & 0,73 \\ 0,5 & 0,73 \end{pmatrix} = \begin{pmatrix} 0,1 & 0,2 \\ 0,25 & 0,2 \end{pmatrix}$$

$$\frac{\partial L}{\partial w^{(1)}} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} \left[\begin{pmatrix} 0 & -0,73 \\ 0 & 1,27 \end{pmatrix} \odot \begin{pmatrix} 0,1 & 0,2 \\ 0,25 & 0,2 \end{pmatrix} \right] =$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -0,146 \\ 0 & 0,254 \end{pmatrix} =$$

$$= \frac{1}{2} \begin{pmatrix} 0 & -0,146 \\ 0 & 0,038 \\ 0 & -0,146 \end{pmatrix} = \begin{pmatrix} 0 & -0,073 \\ 0 & 0,019 \\ 0 & -0,073 \end{pmatrix}$$

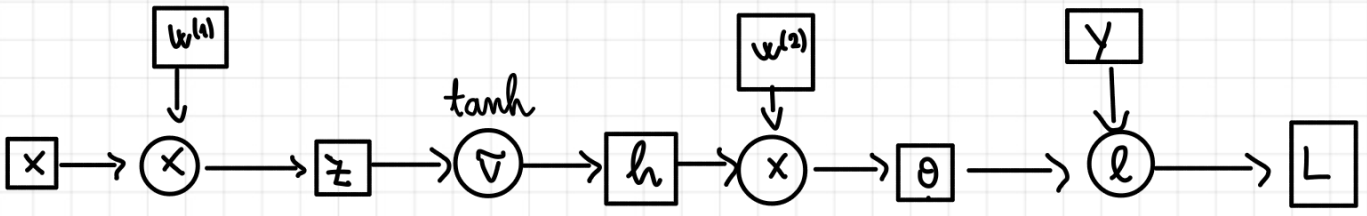
3.

compute the output of attention pooling for scaled dot-product attention.

10. Define an MLP for regression, having 3 inputs, one hidden layer of 2 units, with the hyperbolic tangent $\tanh(z) = \frac{1-e^{-2z}}{1+e^{-2z}}$ activation function, and 2 outputs. We ignore bias terms for simplicity. Moreover, the loss function is the squared error loss, and is denoted as L. Compute symbolically $\frac{\partial L}{\partial W^{(1)}}$. Note that the first intermediate variable is $z = XW^{(1)}$ and not $z = W^{(1)}x$, which means that you will have to deduce the equations in this new setting. Numerical application:

$$W^{(1)} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \\ 1 & 0 \end{bmatrix}, W^{(2)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, X = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}, y = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}.$$

$n = 2 \qquad d = 3 \qquad h = 2 \qquad q = 2$



$$Z = X W^{(1)} \in \mathbb{R}^{2\times 2}$$

$$h = \nabla(Z) \in \mathbb{R}^{2\times 2}$$

$$O = h W^{(2)} \in \mathbb{R}^{2\times 2}$$

$$l(O, y) = \frac{1}{2n} \operatorname{Tr}\left((O-y)^T (O-y)\right)$$

$$\frac{\partial L}{\partial W^{(1)}} = \frac{\partial L}{\partial Z} \cdot \frac{\partial Z}{\partial W^{(1)}} = \frac{\partial L}{\partial Z} \cdot \frac{\partial (X W^{(1)})}{\partial W^{(1)}} = X^T \frac{\partial L}{\partial Z}$$

$$\frac{\partial L}{\partial Z} = \frac{\partial L}{\partial h} \cdot \frac{\partial h}{\partial Z} = \frac{\partial L}{\partial h} \frac{\partial \nabla(Z)}{\partial Z} = \frac{\partial L}{\partial h} \odot \left(1 - \tanh^2(Z)\right)$$

$$\frac{\partial L}{\partial h} = \frac{\partial L}{\partial O} \cdot \frac{\partial O}{\partial h} = \frac{\partial L}{\partial O} \frac{\partial (h W^{(2)})}{\partial h} = \frac{\partial L}{\partial O} W^{(2)T}$$

$$\frac{\partial L}{\partial O} = \frac{1}{2n} \cdot 2 (O-y) = \frac{1}{n} (O-y)$$

$$\frac{\partial L}{\partial W^{(1)}} = X^T \left[\left(\frac{1}{n} (O-y) W^{(2)T} \right) \odot \nabla(Z) \odot (1 - \nabla(Z)) \right]$$

$$Z = X W^{(1)} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 0 & -1 \end{pmatrix}$$

$$\sigma(2) = \frac{1-e^{-4}}{1+e^{-4}} = 0{,}96$$

$$\sigma(-1) = \frac{1-e^{2}}{1+e^{2}} = -0{,}76$$

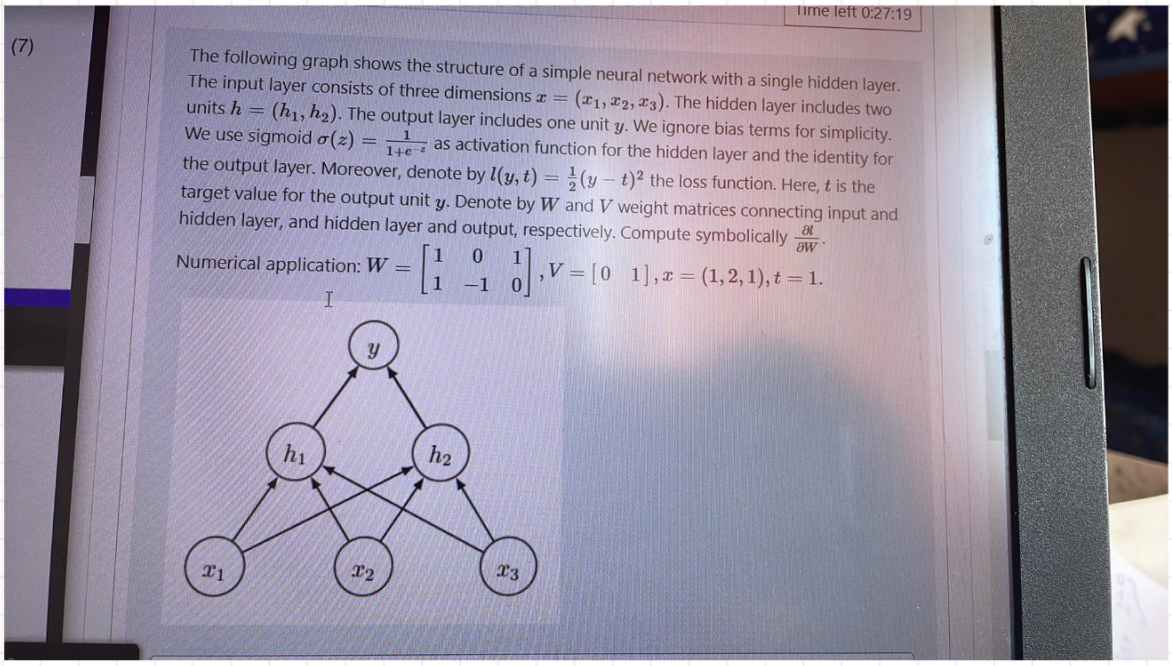
$$\vartheta(0) = 0$$

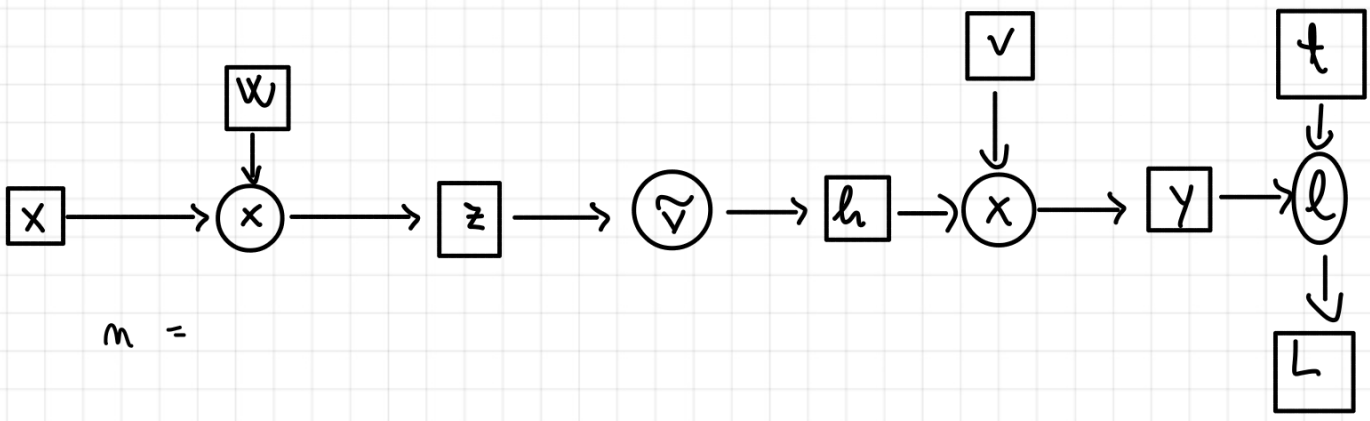
$$h = \begin{pmatrix} 0{,}96 & -0{,}76 \\ 0 & -0{,}76 \end{pmatrix} \qquad \sigma'(z) = \begin{pmatrix} 0{,}07 & 0{,}42 \\ 0 & 0{,}42 \end{pmatrix}$$

$$\frac{\partial L}{\partial z} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} \left[\frac{1}{2} \left[\begin{pmatrix} 0{,}96 & -0{,}76 \\ 0 & -0{,}76 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \right] \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] \odot$$

$$\odot \begin{pmatrix} 0{,}07 & 0{,}42 \\ 0 & 0{,}42 \end{pmatrix} = \begin{pmatrix} -0{,}0028 & -0{,}73 \\ -0{,}005 & -1{,}4 \\ -0{,}0028 & -0{,}73 \end{pmatrix} \cdot \frac{1}{2}$$

4.





$n =$

$z = x W$

$h = \sigma(x W)$

$y = h \cdot V$

$l(y,t) = \frac{1}{2n}(y-t)^2$